Dart Programming Language

String :-

Letter / Digit / Special Character
Punctuation Marks / Emoji

```
void main () {
  String text 1 = "Flutter"; // Double Quote
  String text2 = 'Dart'; // single Quote
  Print (text 1); // Flutter
  Print (text 2); // Dart
}
```

Multi-Line String

```
void main () {
  String text = '''
  Welcome to Dart
  This is multi Line String.
  ''';
  print (text);
}
```

Empty String :-

```
String text = '     ';
```

.length

```
void main () {
  String text = "Dart";
  print (text.length); // 4
}
```

How to store a String in memory?

```
void main() {
  String str = 'Dart';
```

print(str); // Dart

UTF → Unicode Transformation Format

A-Z (65- 90)

a-z (97 – 122)

D → 68 ✓

a → 97 ✓

r → 114 ✓

t → 116 ✓

(01)

print(str.codeUnits)

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	128	64	32	16	8	4	2	1
68		1	0	0	0	1	0	0
97		1	1	0	0	0	0	1
114		1	1	1	0	0	1	0
116		1	1	1	0	1	0	0

68 → 000 000 000 1000 100 ✓ → D

97 → 000 000 000 11 00001 ✓ → a

114 → 000 000 000 111 00 10 ✓ → r

116 → 000 000 000 11 1 0 1 00 ✓ → t

Emoji & Unicode:

Print(emoji.codeUnits); // [55357, 56832]

High Surrogate

Low

Print(emoji.runes);

→ (128512)

Surrogate Pair Formula

Code Point = (highSurrogate − 0x D800) * 0x 400 + (lowSurrogate − 0x DC00) + 0x10000

A−10
B−11
C−12
D−13

$0x\,D800 \rightarrow 16^3 \times 13 + 16^2 \times 8 + 16^1 \times 0 + 16^0 \times 0$

$53248 + 2048$

$= 55296$

$0x\,400 \rightarrow 16^2 \times 4 + 16^1 \times 0 + 16^0 \times 0$

$= 1024$

$0x\,DC00 \rightarrow 16^3 \times 13 + 16^2 \times 12 + 16^1 \times 0 + 16^0 \times 0$

$= 53248 + 3072$

$= 56320$

$0x\,10000 = 16^4 \times 1$

$= 65536$

Code Point = $[55357 - 55296] * 1024 + (56832 - 56320) + 65536$

$= 61 \times 1024 + 512 + 65536$

$= 62464 + 66048$

$= 128512$ ✓

Strings are immutable:-

```
void main () {
String name = "Hello";
print (name);   // Hello
print (name.hashCode);   ________
String ucName = name.toUpperCase();
print (ucName);  // HELLO
print(ucName.hashCode);   ________
print(identical(name, ucName));
                          //false

String sameObj = name ;
print (sameObj);   // Hello
print (sameObj.hashCode);   ________
print (identical(name, sameObj));   //true
```

Escape Characters :

\n → New Line

\t → tab space

\' → single Quote

\" → Double Quote

\\ → Back slash

```
void main () {
print( "Hello \n Dart");
                // Hello
                   Dart
print( "Hello \t Dart");

                //Hello ____ Dart
print('I\'m learning Dart');

          // I'm learning Dart.
```

Raw String

```
void main() {
String raw = r" Raw String : \n won't
               create new Line";
print (raw);
}
```

Properties & Methods :-

```
void main() {
String text = "Dart_Programming";
  print(text.length); //16
  print(text.isEmpty); // false
  print(text.isNotEmpty); // true
  print(text.runtimeType); // String
  print(text[0]); // D
  print(text[3]); // t
  print(text.substring(0, 6)); // Dart_P
  print(text.substring(4)); // _Programming
  print(text.contains("Dart")); // true
  print(text.startsWith("Da")); // true
  print(text.endsWith("ing")); // true
```

trim(), trimRight(), trimLeft()

```
void main() {
String text = "      Dart Programming      ";
  print("|${text}|"); // |      Dart Programming      |
  print("|${text.trimLeft()}|"); // |Dart Programming      |
  print("|${text.trimRight()}|"); // |      Dart Programming|
  print("|${text.trim()}|"); // |Dart Programming|
```

replaceAll()

```
void main() {
String text = "I Love Java";
print(text.replaceAll("Java", "Dart"));
}
```

split() / Join()

```
void main() {
String fruits = "apple, banana, orange";
print(fruits.split(','));
                // ["apple", "banana", "orange"]
}

void main() {
List colors = ['red', 'green', 'blue'];
print(colors.join('-'));
}               // "red-green-blue"
```

compareTo() (0, -1, +1)

```
void main() {
print("apple".compareTo("banana"));
      "orange"                          // +1
                // -1
}
```

padLeft(), padRight()

```
void main() {
print("Dart".padLeft(10, '*'));
        // ******Dart
print("Dart", padRight(10, '-'));
        // Dart------
}
```

Repeat

```
void main() {
print("Hi!..." * 3);
}
        // Hi!... Hi!.... Hi!...
```